Practical 9 and 10 subcode:SECE3660 sub- Image Processing & Steganography Name :sai ERP NO:23SE02CB014

Import Libraries

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Upload the Image

In [2]:
```
image = cv2.imread("woman.png", cv2.IMREAD_GRAYSCALE)

print("Image shape:", image.shape)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")
plt.show()
```

```
Image shape: (640, 640)
```

Original Image

Convert Message to Binary

In [3]:
```
def message_to_binary(message):
    return ''.join(format(ord(i), '08b') for i in message)
```

Encode (Hide Message in Image)

In [4]:
```
def encode_image(image, message):

    message += "###"   # End marker
    binary_message = message_to_binary(message)

    data_index = 0
    data_len = len(binary_message)

    encoded_image = image.copy()

    for row in range(encoded_image.shape[0]):
        for col in range(encoded_image.shape[1]):

            if data_index < data_len:

                pixel = encoded_image[row, col]

                encoded_image[row, col] = (pixel & 254) | int(binary_message[data_index])

                data_index += 1

            else:
                return encoded_image

    return encoded_image
```

Enter Secret Message and Download the Stego Image

In [6]:
```
secret_message = input("Enter secret message: ")

encoded_img = encode_image(image, secret_message)

cv2.imwrite("stego_image.png", encoded_img)

print("Message successfully hidden!")
plt.imshow(encoded_img, cmap='gray')
plt.title("Stego Image (Message Hidden)")
plt.axis("off")
plt.show()
```

```
Message successfully hidden!
```

Stego Image (Message Hidden)

Decode (Extract Hidden Message)

In [7]:
```
def decode_image(image):

    binary_data = ""

    for row in range(image.shape[0]):
        for col in range(image.shape[1]):
            pixel = image[row, col]
            binary_data += str(pixel & 1)

    message = ""

    for i in range(0, len(binary_data), 8):
        byte = binary_data[i:i+8]
        char = chr(int(byte, 2))
        message += char

        if message.endswith("###"):
            break

    return message[:-3]
```

Run Decoder

In [8]:
```
stego_image = cv2.imread("stego_image.png", cv2.IMREAD_GRAYSCALE)

decoded_message = decode_image(stego_image)

print("Decoded Message:", decoded_message)
```

```
Decoded Message: HELLO WORLD
```